

Language Identification Model for Indian Dialects

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Abstract

This project traverses the development of a spoken language identification model for 22 Indian languages using the mHuBERT-147 pretrained model fine-tuned on the badrex/nnti-dataset-full dataset.

1 Introduction

Language identification is a pivotal task in speech processing, which aids in determining the language spoken. Identifying the language precisely is key to speech recognition and translation. The HuBERT model learns from large labeled datasets and can be optimized for language identification. The objective of this project is to improve accuracy and reduce speaker bias. In Task 1 hyperparameter tuning was performed to improve accuracy, and in Task 2 speaker bias was handled through fine-tuning and data-centric augmentations, thereby reducing non-semantic learning. This report explains the methodology used to improve baseline performance in the HuBERT model. Describes the hyperparameter tunings to reduce speaker bias and the analysis results, including t-SNE visualizations, a confusion matrix, and a classification report between baseline and fine-tuned models for closely related languages.

2 Dataset

For this project, we used a dataset from Hugging Face(badrex, 2023). The dataset contained 22 Indian languages and audio samples with speaker identities. Each sample has an audio, a language label, and a speaker identity. All the audio samples were shuffled and resampled to 16 kHz to match the pretrained model (Hsu et al., 2021).

3 Baseline model—Hyperparameter tuning

This project trains and evaluates a model that can recognize the language spoken in an audio clip.

The goal is to identify which language is being spoken based only on the speech signal. The model is trained using the NNTI dataset, which contains audio recordings in different Indian languages. In task01, to reproduce the baseline of 31.96 we first installed all required libraries and downloaded the dataset. Huggingface was used for the implementation. Authentication to the Hugging Face Hub was performed using a Hugging Face access token to enable model downloading and logging. Additionally, an API token for Weights & Biases (WandB) was configured to track training experiments.

For the baseline, we selected the mHuBERT-147 model (Utter Project, 2023). mHuBERT-147 is a compact pretrained speech representation model designed for multilingual speech tasks, offering an unprecedented balance between high performance and parameter efficiency, and is fine-tuned for audio classification, where the output is the predicted language label. (Boito et al., 2024) . Compared to larger multilingual models such as Facebook/mms-300m, mHuBERT is better suited for smaller datasets due to its reduced parameter size and efficient architecture.

Performance and compatibility with the dataset was increased using several parameters: The maximum input audio duration was increased from 7 seconds to 10 seconds to allow the model to capture more linguistic information from each speech sample. During preprocessing, an evaluation step was included to verify that all audio samples were correctly resampled to 16 kHz, which matches the pretraining configuration. Dropout regularization was introduced to improve generalization and reduce overfitting during fine-tuning. Dropout randomly disables a fraction of neurons during training, preventing the model from relying too heavily on specific features(Srivastava et al., 2014).

The following dropout settings were applied:

- config.hidden_dropout = 0.1

- `config.attention_dropout = 0.1`
- `config.activation_dropout = 0.1`
- `config.feat_proj_dropout = 0.1`

The training dataset was prepared using the model's feature extractor. Audio samples were converted into the format expected by the feature extractor, padded to create uniform batch sizes, and corresponding language labels were attached to each sample. The training parameters were then updated to the following:

- `batch_size = 2`
- `gradient_accumulation_steps = 16`
- `num_train_epochs = 7`
- `lr = 0.00002`

Gradient accumulation was used to simulate larger batch sizes for stable optimization (Vaswani et al., 2017). The number of training epochs was updated to 7, and a learning rate was set to 0.00002. The learning rate schedule with warmup steps was used to stabilize optimization during the early stages of training.

To further improve optimization, weight decay and warmup steps were set to :

- Weight decay: 0.04
- Warmup steps: 500

The Maximum accuracy achieved for the baseline model was : 65.3%

4 Data-Centric Augmentation

Using the hyper-tuned HuBERT model (Utter Project, 2023) with improved accuracy, we first imported all base and Hugging Face libraries to enable the transformer model architecture (Wolf et al., 2020). A pretrained feature extractor was utilized to perform normalization, padding, and attention masking on the audio samples data. The dataset (badrex, 2023) was prepared by shuffling and resampling (Utter Project, 2023), after which audio clips are set to 10 seconds to capture better phonetic features. Following the feature extraction step, languages were labeled and sorted, and are then converted into labeled IDs. A data-centric approach was used to enhance model generalization and reduce bias toward speaker identity. During

this step, audio augmentation techniques were performed with the audiomentations library (Jordal, 2020), including pitch shift (min and max semitones set to -4 and 4), Gaussian noise (amplitude between 0.001 and 0.015), and time stretch (between 0.8 and 1.2), all of which simulate various speaking rates and prevent learning speaker identity rather than language. Audio samples were preprocessed with these augmentations, and then the feature extractor determines the maximum length, audio arrays, and sampling rate to generate training samples—speaker IDs and language labels. The data is split into training and validation sets, with augmentations when it is enabled. Dropout is applied for regularization, and the model is loaded for automatic audio classification. Since the audio samples vary in length, the data collator (Hugging Face, 2023) pads inputs and converts labels to tensors. Optimized training parameters and arguments are tuned, and evaluation metrics are set. The model is trained for 7 epochs, achieving approximately 61.3% accuracy. Classification accuracy is then calculated based on the true language labels. Finally, the trained model and feature extractor are pushed to Hugging Face Hub to save them.

5 Evaluation

The baseline model and the improved (fine-tuned) model were loaded from the huggingface hub along with the feature extractor for comparison between them. The dataset (badrex, 2023) was loaded, and the validation split was used for evaluation. Resampled to 16 kHz, (Utter Project, 2023) and labels were stored. Audio was preprocessed using a feature extractor, which returns the audio arrays, sampling rate, and attention masks. Audio Data Collator (Hugging Face, 2023) padded the input and converted the label to tensors. Baseline and Improved model were trained using transformers (Wolf et al., 2020). Torch was used, as the trainer uses PyTorch tensors. The baseline and improved model predictions were made as raw scores; NumPy converts them into class indices. Analyses performed were: 1. Confusion matrix: Constructed to count the number of closely related language predictions that are confused with each other and misclassified due to phonetic or acoustic similarity. 2. Classification report: Calculates individual metrics for each language. 3. t-SNE plots: Explains about the Language clusters and speaker bias

The confusion matrix (Pedregosa et al., 2011)

worked by extracting true labels of languages and converting model scores to labels through the prediction of a baseline, which contains logits for each class. The confusion matrix counted the true positive, false positives, and false negatives.

Classification report (Powers, 2011): The classification report includes metrics like precision, recall, F1 scores, and support, along with accuracy and weighted avg. It works by numeric labels for each language, and metrics are calculated for true positives, false positives, and false negatives.

t-SNE plots: Initiated from the internal hidden layer, which consists of the learned features. The dataset was loaded on a batch basis and evaluated with no gradient for computational efficiency. The final hidden layer was extracted with embeddings that captures phonetic and acoustic information. Mean pooling was performed to get an uniform vector length for all the samples, and dimensionality was reduced to fit a 2D space.

6 Analysis results

The primary source of bias is dataset limitation, as it has only 5 speakers per language, resulting in bias. Learning from this biased data model often chooses shortcut learning to reduce bias. A data-centric augmentation approach was used using audiomentations library(Jordal, 2020). These augmentations reduced speaker bias by disrupting voice, causing variations, and therefore forcing the model to learn phonetic and linguistic features. The key metrics are: The confusion matrices supports that the improved model is learning linguistic traits. The Baseline Model Confusion Matrix shows a higher degree of scattered, somewhat random errors across the board. This happens because when a model memorizes a speaker's voice, it essentially guesses blindly if it fails to recognize that specific acoustic signature in a test sample. The Improved Model Confusion Matrix cleans up much of this random noise. While the baseline accuracy is around 65.3%, the improved model has around 61.3%; there is an accuracy tradeoff as the errors in the improved model are much more concentrated and localized. Instead of random guessing, the improved model makes logical mistakes, meaning it gets confused when it encounters languages that actually sound alike due to shared linguistic roots. Because the augmentation forced the improved model to listen to phonetic patterns rather than speaker pitch, it can be seen it confuses spe-

cific pairs of languages that likely share a language family, has a subtle effect on the accuracy, but improves the speaker bias. For example, true Class 15 is heavily confused with Class 12 (60 times) and Class 14 (61 times). Similarly, true Class 9 is frequently misclassified as Class 18 (38 times) and Class 19 (23 times). Another general example is the high degree of similarity between the Urdu and Hindi languages. Both are very similar which can confuse the model.

The t-SNE plot of both models provide visual proof of our working hypothesis regarding speaker bias. In the Baseline Model t-SNE plot, we can see that the data points for individual languages represented by different colors are present in a broken form in multiple, tight, and separated subclusters. Because our dataset had exactly 5 speakers per language, these distinct subclusters clearly indicate that the baseline model encoded speaker identity. The model has mapped the unique acoustic signatures of those 110 specific individuals instead of languages themselves. On the other hand the improved model section of the t-SNE plot shows that these sub-clusters have merged. The data augmentation techniques, which includes pitch shifting, time stretching, and noise, successfully disrupted the individual voice signatures. Resultantly, the improved model groups the representations into larger, more cohesive single clusters per color, proving it has stopped relying on the shortcut of speaker identity and is instead encoding actual language-level features.

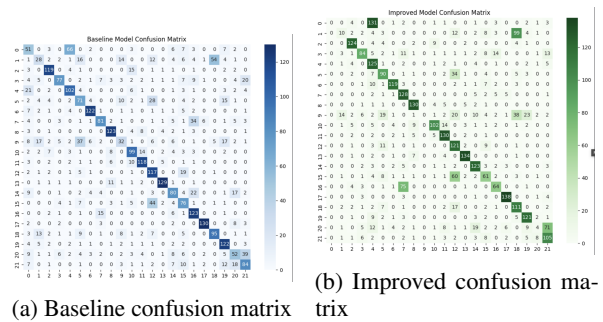


Figure 1: Confusion matrices comparing baseline and improved models

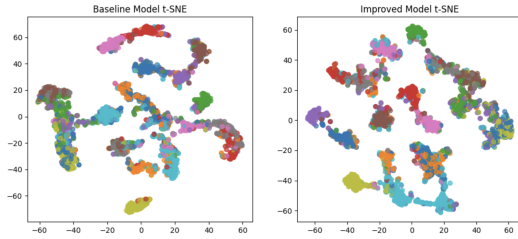


Figure 2: T-SNE Plot.

7 Conclusion

This project developed a good-performing language identification system for 22 Indian languages using the mHuBERT-147 pretrained model. By optimizing the model with given Hugging face dataset ([badrex, 2023](#)), we were able to improve the baseline accuracy from 31.9% to 65.3% by optimization of training parameters and arguments. The main challenge was the speaker bias; To reduce it, the data-centric approach was performed using the audiomentations library ([Jordal, 2020](#)) and included pitch shift, Gaussian noise and time stretch to simulate variations thus helped in reducing speaker bias with an accuracy tradeoff due to the generalization of the model over specific features. Further analysis included generating confusion matrices, which stated the logical confusion between phonetically similar languages like Hindi and Urdu indicating the model encodes linguistic features rather than memorizing speaker identity. Evaluation methods like confusion matrices, classification report and t-SNE plots were used for analysis. However, languages with low F1 scores are still hard to classify and needs a high diverse dataset with more efficient model capacity.

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